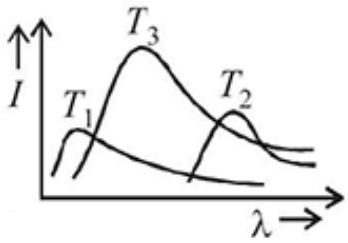
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The plots of intensity versus wavelength for three black bodies at temperatures T_1, T_2 and T_3 respectively are as shown. Their temperature are such that



(A) $T_1 > T_2 > T_3$

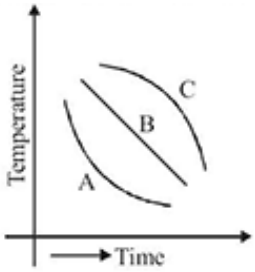
(B) $T_1 > T_3 > T_2$

(C) $T_2 > T_3 > T_1$

(D) $T_3 > T_2 > T_1$

Q.2

A block of steel heated to 100°C is left in a room to cool. Which of the curves shown in fig., represents the correct behaviour?



(A) A

(B) B

(C) C

(D) None of these

Q.3

If λ_m denotes the wavelength at which the radiative emission from a black body at a temperature TK is maximum. Then

(A) $\lambda_m \propto T^{-1}$

(B) $\lambda_m \propto T^4$

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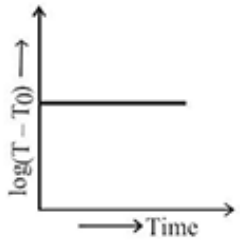
(C) λ_m is independent of T

(D) $\lambda_m \propto T$

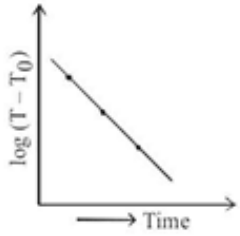
Q.4

Which of the given graphs proves Newton's law of cooling?

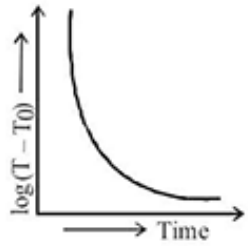
(A)



(B)



(C)



(D) None of these

Q.5

In a room where the temperature is $30^{\circ}\mathrm{C}$, a body cools from $61^{\circ}\mathrm{C}$ to $59^{\circ}\mathrm{C}$ in 4 minutes. Then how much time (in minutes) will be taken by the body to cool from $51^{\circ}\mathrm{C}$ to $49^{\circ}\mathrm{C}$?

(A) 6 min

(B) 4 min

(C) 7 min

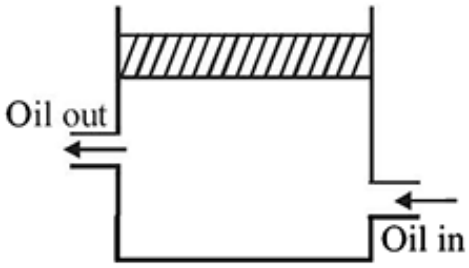
(D) 14 min

Q.6

The top of an insulated cylindrical container is covered by a disc having emissivity 0.6, conductivity 0.167 $\mathrm{WK}^{-1}\ \mathrm{m}^{-1}$ and thickness $1\ \mathrm{cm}$.
The temperature is maintained by circulating oil as shown in figure. Find the radiation loss (in $\mathrm{Jm}^{-2}\mathrm{sec}^{-1}$) to the surrounding in $\mathrm{Jm}^{-2}\ \mathrm{s}^{-1}$ if temperature of the upper surface of the disc is $27^{\circ}\mathrm{C}$ and temperature of

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the surrounding is $127^{\circ}\mathrm{C}$.



(A) $500 Jm^{-2}s^{-1}$

(B) $595 Jm^{-2}s^{-1}$

(C) $615 Jm^{-2}s^{-1}$

(D) $550 Jm^{-2}s^{-1}$

Q.7

The filament of an evacuated light bulb has a length $10\ \mathrm{cm}$, diameter $0.2\ \mathrm{mm}$ and emissivity 0.2. Calculate the power (in watt) it radiates at $2000\ \mathrm{K}$. $\left(\sigma = 5.67 \times 10^{-8}\ \mathrm{W/m^2\ K^4}\right)$

(A) 5W

(B) 1.1W

(C) 6.5W

(D) 11.4W

Q.8

If wavelength of maximum intensity of radiation emitted by sun and moon are $0.5 \times 10^{-6}\ \mathrm{m}$ and $10^{-4}\ \mathrm{m}$ respectively the ratio of their temperature is

(A) $1/10$

(B) $1/50$

(C) 100

(D) 200

Q.9

I wo solid spheres of radi R_1 and R_2 are made of the same material and have similar surfaces. These are raised to the same temperature and then allowed to cool under identical conditions. The ratio of their initial rates of loss of heat are

(A) $(\mathrm{R}_1/\mathrm{R}_2)$

(B) $(\mathrm{R}_2/\mathrm{R}_1)$

(C) $\left(R_1^2/\mathrm{R}_2^2\right)$

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(D) $\left(R_2^2/R_1^2\right)$

Q.10

The wavelength of the radiation emitted by a body depends upon:

(A) the nature of the surface

(B) the area of the surface

(C) the temperature of the surface

(D) all of the above factors

Q.11

A cup of tea cools from $80^\circ C$ to $60^\circ C$ in one minute. The ambient temperature is $30^\circ C$. In cooling from $60^\circ C$ to $50^\circ C$ it will take

(A) 50 seconds

(B) 90 seconds

(C) 60 seconds

(D) 30 seconds

Q.12

A can is taken out from a refigerator at $0^\circ C$. The atmospheric temperature is $25^\circ C$. If t_1 is the time taken to heat from $0^\circ C$ to $5^\circ C$ and t_2 is the time taken from $10^\circ C$ to $15^\circ C$, then

(A) $t_1 > t_2$

(B) $t_1 < t_2$

(C) $t_1 = t_2$

(D) there is no relation

Q.13

The temperature of perfect black body is $727^\circ C$ and its area is $0.1\ m^2$. If Stefan's constant is $5.67 watt/m^2 sK^{-4}$, then heat radiated by it in 1 minute is

(A) $8100 cal$

(B) $81000 cal$

(C) $810 cal$

(D) $81 cal$

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Q.14

A rod $1\ \mathrm{m}$ long and made of material of thermal conductivity $420\ \mathrm{W/m/K}$ has one of its ends in melting ice and the other end in boiling water. If its area of cross-section is $10\ \mathrm{cm}^2$, the amount of ice that melts in 1 minute is

(A) $0.125\ \mathrm{g}$

(B) $75\ \mathrm{g}$

(C) $7.5\ \mathrm{g}$

(D) $450\ \mathrm{g}$

Q.15

The coefficient of real expansion of mercury is $18\times\ 10^{-5}/^{\circ}\mathrm{C}$. A thermometer has a bulb of volume $10^{-6}\ \mathrm{m}^3$ and the cross-sectional area of the stem is $0.002\ \mathrm{cm}^2$. The bulb is completely filled with mercury when the temperature is $0^{\circ}\mathrm{C}$. When the temperature rises to $100^{\circ}\mathrm{C}$, the length of the mercury column in the stem will be

(A) $9\ \mathrm{cm}$

(B) $18\ \mathrm{cm}$

(C) $9\ \mathrm{mm}$

(D) $18\ \mathrm{mm}$

Q.16

Suppose you come across old scientific notes that describe a temperature scale called Z on which the boiling point of water is $65.0^{\circ}\mathrm{Z}$ and the freezing point is $-14.0^{\circ}\mathrm{Z}$.
To what temperature on the Fahrenheit scale would a temperature of $T = -98.0^{\circ}\mathrm{Z}$ correspond? Assume that the Z scale is linear; that is, the size of a Z degree is the same everywhere on the Z scale.

(A) -159°F

(B) -139°F

(C) -149°F

(D) -129°F

Q.17

Two ideal gas thermometers A and B use oxygen and hydrogen, respectively. The following observations are made:
\begin{array}{|c|c|c|} \ \hline \text { Temperature } & \begin{array}{c} \ \text { Pressure } \\ \ \text { thermometer } A \ \end{array} & \begin{array}{c} \ \text { Pressure } \\ \ \text { thermometer B } \ \end{array} \\ \ \hline \text { Triple point of water } & 1.250 \times 10^{5} \mathrm{~Pa} & 0.200 \times 10^{5} \mathrm{~Pa} \\ \ \hline \begin{array}{c} \ \text { Normal melting point of } \\ \ \text { sulphur } \end{array} & & \end{array}

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& 1.797 \times 10^{5} \mathrm{~Pa} & 0.287 \times 10^{5} \mathrm{~Pa} \\ \ \hline \ \end{array}
(a) What is the absolute temperature of normal melting point of sulphur as read by thermometer A and B ?

(A) For thermometer A- 392.46 K, For thermometer B- 391.98 K

(B) For thermometer A- 393.46 K, For thermometer B- 391.98 K

(C) For thermometer A- 392.46 K, For thermometer B- 393.98 K

(D) For thermometer A- 395.46 K, For thermometer B- 396.98 K

Q.18

What will be the values of the following temperatures on the Kelvin scale: (a) $37^{\circ} \mathrm{C}$, (b) $80^{\circ} \mathrm{F}$, (c) $-196^{\circ} \mathrm{C}$?

(A) 310 K,299.66 K, 77 K

(B) 330 K,305 K, 79K

(C) 320 K,299.66 K, 78 K

(D) 310 K,305 K, 77 K

Q.19

A body cools down from 60° Cto 55° C in 30 s. Using Newton's law of cooling, calculate the time taken by same body to cool down from $55^{\circ} \mathrm{C}$ to $50^{\circ} \mathrm{C}$. Assume that the temperature of surrounding is $45^{\circ} \mathrm{C}$.

(A) 58.28 s

(B) 51.28 s

(C) 53 s

(D) 59 s

Q.20

An electric heater of surface area $200 \mathrm{~cm}^{2}$ emits radiant energy of $60 \mathrm{~kJ}$ at time interval of $1 \mathrm{~min}$. Determine its emissive power. If its emissivity be 0.45, what would be the radiant energy emitted by a black body in one hour, identical to the electrical heater in all respects?

(A) 4000 Kj

(B) 3300 Kj

(C) 8000 Kj

(D) 3170 Kj

Answers & Solutions

Q.1 Answer:

$T_{1}>T_{3}>T_{2}$

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Solution:

According to Wien's law $\lambda_m \propto \frac{1}{T}$ and from the figure $(\lambda_m)_1 < (\lambda_m)_3 < (\lambda_m)_2$ therefore $T_1 > T_3 > T_2$

Q.2 Answer:

A

Solution:

According to Newton's law of cooling if temperature difference between body \& surrounding is large, then rate of cooling is also fast hence curve A shows correct behaviour.

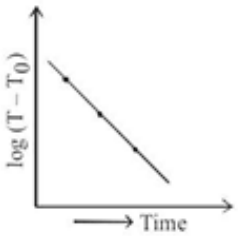
Q.3 Answer:

$\lambda_m \propto T^{-1}$

Solution:

From Wein's displacement law $\lambda_m T =$ constant
$\Rightarrow \lambda_m \propto T^{-1}$

Q.4 Answer:



Solution:

When hot water temperature (T) and surrounding temparature (T_0) readings are noted, and $\log(T - T_0)$ is plotted versus time, we get a straight line having a negative slope; as a proof of Newton's law of cooling.

Q.5 Answer:

6 min

Solution:

From Newtwon's law of cooling
$\frac{\mathrm{dQ}}{\mathrm{dt}} = -\mathrm{KA}\frac{\mathrm{dT}}{\mathrm{dx}}$
Area of cross-section A and thickness dx is the same. Also $\mathrm{dQ} = \mathrm{mCd}\theta$
Thus in first case $\frac{\mathrm{m}\times\mathrm{C}\times(61^\circ - 59^\circ)}{4} = \frac{-\mathrm{KA}}{\mathrm{dx}}\left[\left(\frac{61^\circ + 51^\circ}{2}\right) - 30^\circ\right]$
In second case, $\frac{m\times C\times(51^\circ - 49^\circ)}{t} = \frac{-KA}{dx}\left[\left(\frac{51^\circ + 49^\circ}{2}\right) - 30^\circ\right]$
Dividing equation (i) by equation (ii)

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$\frac{t}{4}=\frac{30}{20}$
or $t=6$ **minutes.**

Q.6 Answer:

$595 J m^{-2} s^{-1}$

Solution:

The rate of heat loss per unit area due to radiation $=\in \sigma\left(T^4-T_0^4\right)$
$=0.6 \times 5.67 \times 10^{-8}\left[(400)^4-(300)^4\right]=595 \mathrm{Jm}^{-2} \mathrm{~s}^{-1}$

Q.7 Answer:

11.4W

Solution:

$\ell=10 \mathrm{~cm}=0.1 \mathrm{~m}, \mathrm{~d}=0.2 \mathrm{~mm}$

$$r=0.1 \mathrm{~mm}=1 \times 10^{-4} \mathrm{~m},$$

$$e=0.2, \mathrm{~T}=2000 \mathrm{~K}, \sigma=5.67 \times 10^{-8} \mathrm{~W} / \mathrm{m}^2 \mathrm{~K}^4$$

According to stefan's law of radiation, rate of emission of heat for an ordinary body,

$E=\sigma A e T^4=\sigma(2 \pi r \ell) e T^4$
$=5.67 \times 10^{-8} \times 2 \times 3.14 \times 1 \times 10^{-4} \times 0.1 \times 0.2 \times(2000)^4$
$=11.4 \mathrm{~W}$

$\therefore$ **Power radiated by the filament** $=11.4 \mathrm{~W}[\mathrm{~A}=2 \pi r \ell]$

Q.8 Answer:

200

Solution:

wavelength of radiation emitted from Sun $\lambda_s=0.5 \times 10^{-6} \mathrm{~m}$
wavelength of the radiation emitted from the moon $\lambda_m=10^{-4}$
Wein's law
$T \lambda=$ **constant**
$\frac{T_s}{T_m}=\frac{\lambda_m}{\lambda_s}=\frac{10^{-4}}{0.5 \times 10^{-6}}=200$

Q.9 Answer:

$\left(R_1^2 / R_2^2\right)$

Solution:

The ratio of heat loss is
$\frac{H_1}{H_2}=\frac{\left(\frac{dQ}{dt}\right)_1}{\left(\frac{dQ}{dt}\right)_2}$
If the spheres are raised to the same temperature than allowed to cool

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$\frac{H_1}{H_2} = \frac{KA_1}{KA_2}$
$= \frac{R_1^2}{R_2^2}$

Q.10 Answer:

the temperature of the surface

Solution:

According to Wien's law $\lambda \propto \frac{1}{T}$
i.e., it depends on the temperature of the surface.

Q.11 Answer:

50 seconds

Solution:

$t = \frac{1}{K} \log_{10} \left(\frac{\theta_1 - \theta_2}{\theta_2 - \theta_0} \right)$
$\theta_1 = 80°$, **surrounding term** $\theta_0 = 30°C$ **and** $\theta_2 = 50°C$
$60 = \frac{1}{K} \log_{10} \left(\frac{80-30}{60-30} \right) \quad 60 = \frac{1}{K} \log_{10} \left(\frac{5}{3} \right) \quad \ldots\ldots$ **(i)**
Let the the of cooling of the gas from temperature $60°C$ **to** $50°C$
$t' = \frac{1}{K} \log_{10} \left(\frac{60-30}{50-30} \right) = \frac{1}{K} \log_{10} \left(\frac{3}{2} \right)$
$\frac{t'}{60} = \frac{\log_{10}\left(\frac{3}{2}\right)}{\log_{10}\left(\frac{5}{3}\right)}$
$t' = 50\text{sec}$

Q.12 Answer:

$t_1 < t_2$

Solution:

Correct option is (b) $t_1 < t_2$
Sir Newton's law of heating states that rate of change of temperature of a object is proportional to the difference in temperatures of the surrounding and the object.
In the given cases, when initial temperature of can is $0°C$**, the difference in temperatures of can and surrounding is** $25 - 0 = 25°C$.
When initial temperature of can is $10°C$**, the difference in temperatures of can and surrounding is** $25 - 10 = 15°C$ **therefore the rate of heating will be higher, when temperature difference is** $25°C$
i.e. time t_1 **will be smaller than** $\$\$t_{-}\{2\}, t_1 < t_2$

Q.13 Answer:

81000cal

Solution:

$Q = A\sigma t T^4$
$\frac{0.1 \times 5.67 \times 10^{-8} \times 60 \times (1000)^4}{4.2} \text{cal}$

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$= 81000\text{cal}$

Q.14 Answer:

$7.5\ \text{g}$

Solution:

Given $k = 420\ \text{W/m/K}, A = 10\ \text{cm}^2 = 10 \times 10^{-4}\ \text{m}^2$
$= 10^{-3}\ \text{m}^2, \theta_1 = 100^\circ\text{C}, t_2 = 0^\circ\text{C}, t = 1$ **minute** $= 60\ \text{s}$
and $l = 1\ \text{m}$**. Using these values in**
$Q = \frac{kA(\theta_1 - \theta_2)t}{l}$
we get $Q = 2520\ \text{J} = \frac{2520}{4.2} = 600\text{cal}$**. Therefore,**
mass of ice melted is
$m = \frac{Q}{L} = \frac{600\text{cal}}{80\text{cal/g}} = 7.5\ \text{g}$
Hence the correct choice is (c).

Q.15 Answer:

$9\ \text{cm}$

Solution:

Increase in volume of mercury when the temperature increases by 100°C **is** $v = \gamma_r V \Delta T =$
$\left(18 \times 10^{-5}\right) \times \left(10^{-6}\right) \times 100 = 18 \times 10^{-9}\ \text{m}^3$
Now
$v = Ah$
$h = \frac{v}{A}$
or $= \frac{\left(18 \times 10^{-9}\ \text{m}^3\right)}{\left(0.002 \times 10^{-4}\ \text{m}^2\right)} = 0.09\ \text{m} = 9\ \text{cm}$
Hence the correct choice is (a).

Q.16 Answer:

-159°F

Solution:

For making a conversion factor between two temperature scales we use two known (benchmark) temperatures, such as the boiling and freezing points of water.
The number of degrees between the known temperatures of one scale is equivalent to the number of degrees between them on the other scale.
The unknown temperature T **on the** Z **scale** $\left(T = -98.0^\circ\text{Z}\right)$ **closer to the freezing point** $\left(-14.0^\circ\text{Z}\right)$ **than to the boiling poin** $\left(65.0^\circ Z\right)$**, we use the freezing point as reference point. The difference of** T **with respect to the freezing point** $\left(-14.0^\circ\text{Z}\right)$**.**
$\Delta T = \left(-14.0^\circ Z\right) - \left(-98.0^\circ Z\right) = 84.0^\circ Z$
Next, we set up a conversion factor between the Z **and Fahrenheit scales to convert this difference. To do so, we use both known temperatures on the** Z **scale and the corresponding temperatures on the Fahrenheit scale.**

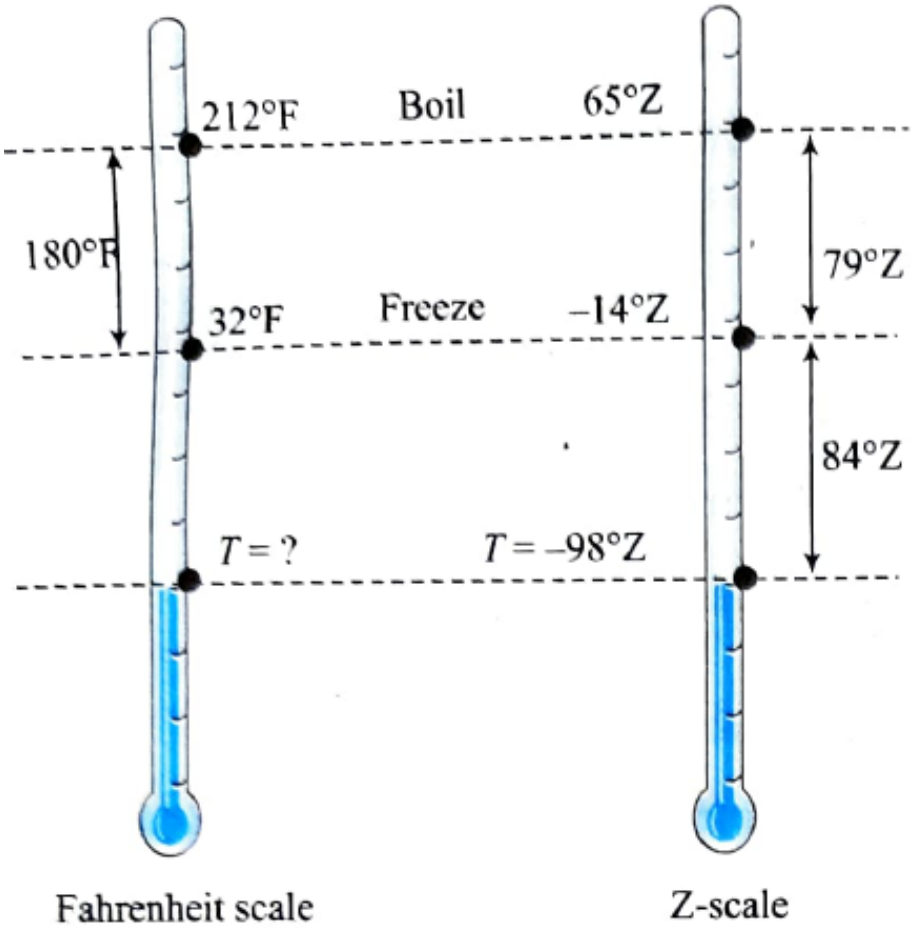
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On the Z scale, the difference between the boiling and freezing points is

$$\Delta T_Z = (65.0^\circ\text{Z}) - (-14.0^\circ\text{Z}) = 79.0^\circ\text{Z}$$

On the Fahrenheit scale, the difference between the boiling and freezing points is $\Delta T_F = (212^\circ\text{F}) - (32^\circ\text{F}) = 180^\circ\text{F}.$

Thus, a temperature difference of 79.0°Z **is equivalent to a temperature difference of** 180°F **(figure), and we can use the ratio** $(180^\circ\text{F}) / (79.0^\circ\text{Z})$ **as our conversion factor.**



Now, since T **is below the freezing point by** 84.0°Z **on** Z **scale, hence it must also be below the freezing point on the Fahrenheit scale by** $(84.0^\circ\text{Z})\ \frac{180^\circ\text{F}}{79.0^\circ\text{Z}} = 191^\circ\text{F}.$

Since the freezing point is at 32.0°F**, we have**

$$T_F = 32^\circ\text{F} - 191^\circ\text{F} = -159^\circ\text{F}$$

Q.17 Answer:

For thermometer A- 392.46 K, For thermometer B- 391.98 K

Solution:

(a) For thermometer $A, T_{\text{tr}} = 273.16\ \text{K}, P_{\text{tr}} = 1.250 \times 10^5\ \text{Pa}$

We have $T = \frac{P}{P_{\text{tr}}} \times T_{\text{tr}} = \frac{1.797 \times 10^5}{1.250 \times 10^5} \times 273 = 392.46\ \text{K}$

For thermometer $B, T_{\text{tr}} = 273.16\ \text{K}, \quad P_{\text{tr}} = 0.200 \times 10^5\ \text{Pa}$

We have $T = \frac{P}{P_{\text{tr}}} \times t_{\text{tr}} = \frac{0.287 \times 10^5 \times 273.16}{0.200 \times 10^5} = 391.98\ \text{K}$

(b) The slight difference in the temperatures as read by two thermometers is due to the fact that oxygen and hydrogen do not behave like an ideal gas.

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Q.18 Answer:

310 K,299.66 K, 77 K

Solution:

(a) Temperature on Kelvin scale T_{K} is related to temperature T_{C} on Celsius scale as

$T_{\mathrm{K}} = T_{\mathrm{c}} + 273$

or $T_K = 37 + 273 = 310\ \mathrm{K}$

(b) Temperatures T_K on Kelvin scale and T_p on Fahrenheit scale are related as

$\frac{T_{\mathrm{K}}-273}{373-273} = \frac{T_{\mathrm{F}}-32}{212-32}$

or $\frac{T_K-273}{100} = \frac{T_{\mathrm{F}}-32}{180}$

or $T_{\mathrm{K}} = \frac{5}{9}(T_{\mathrm{F}} - 32) + 273$

here $T_{\mathrm{F}} = 80^{\circ}\mathrm{F}$**; thus**

$T_{\mathrm{K}} = \frac{5}{9}(80 - 32) + 273 = 299.66\ \mathrm{K}$

(c) Again from relation used in part (a)

$T_{\mathrm{K}} = T_{\mathrm{C}} + 273 = -196 + 273 = 77\ \mathrm{K}$

Q.19 Answer:

51.28 s

Solution:

Assume that a body cools down from temperature θ_i to θ_1 In t seconds, and θ_c is the temperature of surroundings. Applying Newton's law of cooling, According to Newton's law of cooling

$\frac{d\theta}{dt} = -K(\theta - \theta_0)\,; \quad \theta_0 = 45^{\circ}\mathrm{C}; \quad \int \frac{d\theta}{\theta-45} = -k\int dt$ **(K is constant)**

from $t = 0$ **s to** $t = 30$ **s,** θ **changes from** $60^{\circ}\mathrm{C}$ **to** $55^{\circ}\mathrm{C}$. $\int_{55}^{60} \frac{d\theta}{\theta-45} = -K\int_{30}^{0} dt$

$\Rightarrow \quad \ln\left(\frac{50-45}{60-45}\right) = -K(30-0)$ **...(i)**

$\int_{55}^{5} \frac{d\theta}{\theta-45} = -K\int_{30}^{\prime} dt$

$\Rightarrow \quad \ln\left(\frac{50-45}{55-45}\right) = -K(t-30)$ **...(ii)**

Divide Eq. (ii) by Eq. (i) to get

$$\frac{1-30}{30} = \frac{\ln\frac{50-45}{55-45}}{\ln\frac{55-45}{60-45}} \Rightarrow \frac{t-30}{30} = \frac{\ln 2}{\ln 3/2}$$

$$\Rightarrow \quad t = 30\left(1 + \frac{\ln 2}{\ln 3/2}\right) = 81.28\ \mathrm{s}$$

Time taken to cool down from $\theta = 55^{\circ}\mathrm{C}$ **to** $\theta = 50^{\circ}\mathrm{C}$ **is** $(t-30) = (81.28-30) = 51.28$ s

Q.20 Answer:

8000 Kj

Solution:

Given $\Delta Q = 60\ \mathrm{kJ}; \quad A = 200 \times 10^{-4}\ \mathrm{m}^2$ **and** $\Delta t = 60$ s

Using, emissive power $E = \left(\frac{\Delta Q}{\Delta t}\right)\frac{1}{A}$**, we have**

$E = \frac{60\ \mathrm{kJ}}{200\times10^{-4}\ \mathrm{m}^2\times60\ \mathrm{s}} = 50\ \mathrm{kW/m}^2$

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$\therefore$ **Emissivity (of a surface)**

$e = \frac{\text{emissive power (of that surface)}}{\text{emissive power (of a black body)}}$

$\therefore$ **Emissive power of a black body**

$= \frac{50}{0.45}\ \text{kW/m}^2 = 111.11\ \text{kW/m}^2$

$\therefore$ **Heat radiated** $\Delta Q = EA\Delta t$

$= \left(111.11 \frac{\text{kW}}{\text{m}^2}\right)\left(200 \times 10^{-4}\ \text{m}^2\right)(60 \times 60\ \text{s}) = 8000\ \text{kJ}$

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